

4	(a) (i) frequency at which object is made to vibrate/oscillate	B1	[1]
	(ii) frequency at which object vibrates when free to do so	B1	[1]
	(iii) maximum amplitude of vibration of oscillating body when forced frequency equals natural frequency (of vibration)	B1 B1	[2]
	(b) e.g. vibration of quartz/piezoelectric crystal (<i>what is vibrating</i>) <i>either</i> for accurate timing	M1	
	<i>or</i> maximise amplitude of ultrasound waves (<i>why it is useful</i>)	A1	[2]
	(c) e.g. vibrating metal panels (<i>what is vibrating</i>) <i>either</i> place strengthening struts across the panel	M1	
	<i>or</i> change shape/area of panel (<i>how it is reduced</i>)	A1	[2]